

Johann Römer

Independent Researcher

Marburg, Germany

[Contact via Zenodo: 10.5281/zenodo.17472834]

1970-01-01

[RECIPIENT NAME]

**Subject: arXiv Endorsement Request / Collaboration Inquiry — UTAC Framework
(Threshold Dynamics)**

Dear Professor [LAST NAME],

I am writing to request your endorsement for submitting my manuscript "*Domain-Specific Universality in Threshold Activation Criticality: A Multi-Attractor Framework*" to arXiv (physics.data-an or cond-mat.stat-mech).

[PERSONALIZATION BLOCK — Choose relevant option:]

[OPTION A — Their Research Aligns:]

Your research on [SPECIFIC TOPIC, e.g., "early warning signals for climate tipping points" / "emergent abilities in large language models" / "critical phenomena in complex systems"] directly intersects with my findings. Specifically, your [PAPER TITLE, YEAR] demonstrated [KEY RESULT], which aligns remarkably with the UTAC framework's prediction that [CONNECTION TO THEIR WORK].

[OPTION B — Methodological Connection:]

Your expertise in [RENORMALIZATION GROUP THEORY / STATISTICAL MECHANICS / COMPLEX SYSTEMS MODELING] makes you ideally positioned to evaluate this work. The UTAC framework extends Wilson-Kogut RG theory to empirical threshold systems, revealing domain-specific universality classes rather than strict convergence.

[OPTION C — Shared Interest in Applications:]

Given your work on [AI SAFETY / CLIMATE TIPPING POINTS / CONSCIOUSNESS / NEURODEGENERATIVE DISEASES], you may find particular interest in the UTAC framework's applications to [RELEVANT DOMAIN]. The discovery that [KEY FINDING RELEVANT TO THEM] has direct implications for [THEIR RESEARCH AREA].

The manuscript presents empirical analysis of 78 threshold systems across 5 scientific domains (Informational, Biological, Climate, Neurodegenerative, Geophysical), quantifying transition steepness via parameter β . Key findings include:

- **Statistical validation** of domain-specific β -clustering (ANOVA: $F(4, 73) = 185.3$, $p < 10^{-20}$, $\eta^2 = 0.91$)

- Identification of the **Computational Criticality Universality Class** (CCUC) at RG fixed point $\beta \approx 4.2$
- Discovery of **Golden Ratio hierarchical scaling**: $\beta_n \approx \Phi^{n/3}$ validated with $\approx 8\%$ error
- Applications to AI emergence (LLMs), consciousness (neuronal avalanches), and climate tipping points

The full manuscript (25 pages), dataset (78 systems, CSV format), and analysis code are publicly available on Zenodo (DOI: 10.5281/zenodo.17472834).

Request: I would be honored if you could provide arXiv endorsement for physics.data-an (primary) or cond-mat.stat-mech (secondary). I am also open to discussing potential collaboration or co-authorship if the work aligns with your research interests.

[OPTIONAL — If seeking journal submission advice:]

Alternatively, if you believe this work would benefit from further development before arXiv submission, I would greatly value your guidance on next steps. I am targeting *Chaos* or *Physical Review E* for peer review.

Thank you for considering this request. I have attached the executive summary (2 pages) for quick reference, with the full manuscript available at the Zenodo link above. I am happy to provide any additional information or clarification you may need.

Sincerely,

Johann Römer

Independent Researcher

Zenodo: <https://doi.org/10.5281/zenodo.17472834>

Attachments:

- Executive Summary (2 pages)
- Link to Full Manuscript & Dataset: <https://doi.org/10.5281/zenodo.17472834>

Quick Access:

Zenodo Repository: <https://doi.org/10.5281/zenodo.17472834>

Main Paper (PDF): [Direct link after Zenodo upload]

Executive Summary (PDF): [Direct link after Zenodo upload]